Chapter 5
SECOND DEGREE EQUATIONS

1. Which of the following are the solutions of the second degree equation $x^2 - 5x + 6 = 0$.

   A) 2, 3 B) -2, -3

   D) 1, 6 D) 5, -6

2. If one solution of second degree equation $x^2 - 7x + k = 0$ is 3, then what is the value of k?

   A) 10 B) 12

   C) 18 D) 21

3. $x^2 + 4x - 5 = 0$ is a second degree equation, then which of the following is a solution?

   A) 6 B) 5

   C) 1 D) -1

4. Sum of two numbers is 4 and their product is 2, then the numbers are .....

   A) $1+\sqrt{2},\ 1-\sqrt{2}$ B) $\sqrt{2}+1,\ \sqrt{2}-1$

   C) $2+\sqrt{2},\ 2-\sqrt{2}$ D) $\sqrt{2}+2,\ \sqrt{2}-2$

5. Read the following statements given below.

   Statement 1 : The sum of first 10 terms of the arithmatic sequence 99, 97, 95, is 900.

   Statement 2 : The sum of first 90 terms of the arithmatic sequence 99, 97, 95, is 900.

   Which of the following is true?

   A) Statement 1 is true, statement 2 is false

   B) Statement 1 is false, statement 2 is true

   C) Both statements are true

   D) Both statements are false

6. If x is a natural number then,
   i) Which number is to be added to $x^2 + 8x$ to get a perfect square?
   ii) If $x^2 + px + 36$ is a perfect square, then find the value of p?
   iii) If $x^2 + px + q$ is a perfect square, then prove that $p^2 = 4q$.

7. The product of 2 consecutive mutiples of 6 is 352.
   i) By taking one multiple of 6 as x, then write the given statement as an algebraic expression.
   ii) Find the numbers

8. The sum of consecutive natural numbers starting from 1 is 465. How many natural numbers were added?

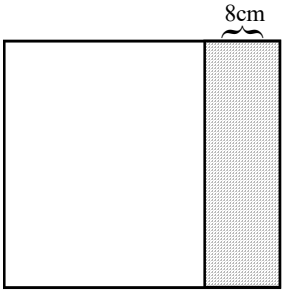
9. The sum of squares of 2 consecutive natural number is 365. Find the numbers.

10. i) If x is an integer, the value of the polynomial $x^2 + 5x$ is 14. By taking x as the same number, then find the value of the polynomial $x^2 + 5x - 14$?
    ii) Find the solutions of the equation $x^2 + 5x - 14 = 0$.

11. The sum of square of a natural number and its six times is 160.
    i) By taking the natural as x , write a second degree equation?
    ii) Find the number.

12. How many terms of the arithmatic sequence 4, 6, 8, ... must be added to get 460?

13. From a square, a rectangle of width 8 cm is cut out. The area of the remaining rectangle is 84 cm^2.
    i) By taking the length of side of the square as x, write the length of the side of rectangle.
    ii) Find the length of sides of the square.



14. How many terms of the arithmatic sequence 59,57,55,..... must be added to get the sum 500?

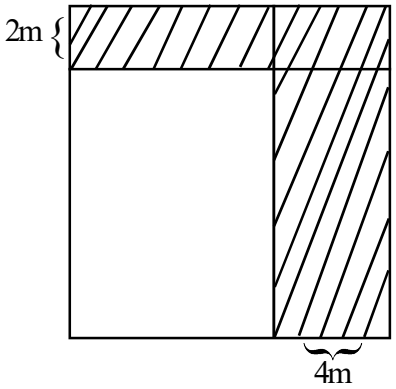
15. A wire of length 80 cm is bent to make a rectangle. The area of the rectangle is 384 cm^2. Calculate the length and breadth of the rectangle?

16. Keerthi is 6 years older than Anu. After 10 years the product of ages of Keerthi and Anu is 247.
    i) By taking the present age of Anu as 'x' years, write the present age of Keerthi.
    ii) Calculate the present ages of Keerthi and Anu?

17. From a rectangle of sides 25 cm & 15 cm a small square is cut out. The area of the remaining part is 326 cm^2. Calculate the length of the side of square which is cut out from the rectangle.

18. The difference of two numbers is 12 and their product is 448. Find the numbers.

19. In a square ground, two roads of width 2m and 4m are constructed as shown below. The area of the remaining part is 575m^2. Calculate the length and breadth of the ground after constructing the roads.



20. The perimeter and area of a rectangle are 80cm and 204 cm^2 respectively. Calculate the length and breadth of the rectangle?

Chapter 5

SECOND DEGREE EQUATIONS

ANSWERS

1. A

2. B

3. C

4. C

5. C

6. i) 16

   ii) 12

   iii) $\left(\frac{P}{2}\right)^2 = q$

   $P^2 = 4q$

7. i) $x\ (x+6) = 352$

   $x^2 + 6x = 352$

   ii) $x^2 + 6x + 9 = 352 + 9 = 361$

   $(x+3)^2 = 19^2$

   $x = 19 - 3 = 16$

   Numbers 16, 22

8. $1+2+3+.....+n = 465$

   $\frac{n(n+1)}{2} = 465$

   $n^2 + n = 930$

   $n^2 + n + \frac{1}{4} = 930 + \frac{1}{4}$

   $\left(n + \frac{1}{2}\right)^2 = \frac{3721}{4} = \left(\frac{61}{2}\right)^2$

$$n+\frac{1}{2}=\frac{61}{2}$$

n = 30

9. $n^2+(n+1)^2=365$

$$2n^2+2n+1=365$$

$$2n^2+2n=364$$

$$n^2+n=182$$

$$n^2+n+\frac{1}{4}=182+\frac{1}{4}$$

$$\left(n+\frac{1}{2}\right)^2=\frac{729}{4}=\left(\frac{27}{2}\right)^2$$

$$n+\frac{1}{2}=\frac{27}{2}$$

Numbers 13, 14

10. i) 0

ii) $x^2+5x-14=0$

$$(x-2)\times(x+7)=0$$

$$x=2,\ -7$$

11. i) $x^2+6x=160$

ii) $x^2+6x+9=160+9=169$

$$(x+3)^2=13^2$$

$$x=10$$

12. $x_n=2n+2$

$$S_n=\frac{n}{2}(4+2n+2)$$

$= \frac{n}{2}(2n+6)$

$= n^2 + 3n$

$n^2 + 3n = 460$

$n^2 + 3n + \frac{9}{4} = 460 + \frac{9}{4}$

$\left(n + \frac{3}{2}\right)^2 = \frac{1849}{4} = \left(\frac{43}{2}\right)^2$

$n + \frac{3}{2} = \frac{43}{2}$

$n = 20$

13. i) $x,\ x-8$

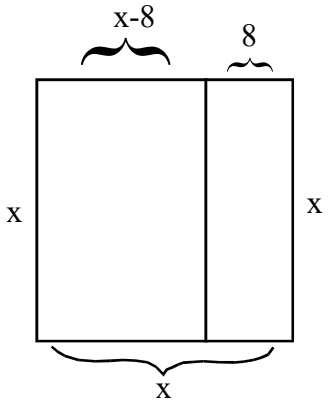
ii) $x(x-8) = 84$

$x^2 - 8x = 84$

$x^2 - 8x + 16 = 100$

$(x-4)^2 = 10^2$

$x = 14$



14. $x_n = -2n + 61$

$S_n = \frac{n}{2}\ [59 + -2n + 61]$

$= \frac{n}{2}\ [-2n + 120]$

$= -n^2 + 60\ n$

$-n^2 + 60n = 500$

$n^2 - 60n = -500$

$n^2 - 60n + 900 = -500 + 900 = 400$

$(n-30)^2 = (\pm 20)^2$

n = 20 +30 or n=30-20

= 50 = 10

15. Length = x, Breadth = 40-x

$x(40-x)=384$

$40x-x^2=384$

$x^2-40x=-384$

$x^2-40x+400=400-384=16$

$(x-20)^2=4^2$

$x=24$

Length = 24 cm Breadth = 16 cm

Or

Length = 20+x Breadth = 20-x

$(20+x)\ (20-x)=384$

$400-x^2=384$

$x^2=400-384=16$

$x=4$

Length = 24 cm, Breadth = 16 cm

16. i) $x+6$

ii) $(x+10)\ (x+16)=247$

$x^2+26x+160=247$

$x^2+26x+169=256$

$(x+13)^2=16^2$

$x=16-13=3$

Age of Anu = 3

Age of Keerthi = 9

17. Let 'x' be the side of the square

$x^2=25x15-326=49$

$x=7$

18. Numbers $x,\ x+12$

$(x+12)x = 448$

$x^2 + 12x = 448$

$x^2 + 12x + 36 = 484$

$(x+6)^2 = (\pm 22)^2$

$x = 22 - 6 = 16$ *or* $x = -22 - 6 = -28$

Numbers **16, 28**

& -28, -16

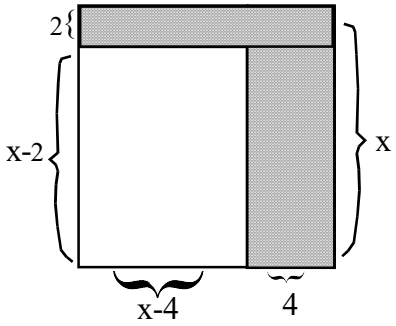
19. $(x-2)\ (x-4) = 575$

$x^2 - 6x + 8 = 575$

$x^2 - 6x + 9 = 576$

$(x-3)^2 = 24^2$

$x = 27$



Length = 25cm, Breadth = 23cm

20. 2 (Length + Breadth) = 80, Length + Breadth = 40

Length = x, Breadth = 40- x

$x\ (40-x) = 204$

$40x - x^2 = 204$

$x^2 - 40x = -204$

$x^2 - 40x + 400 = -204 + 400 = 196$

$(x-20)^2 = 14^2$

$x = 20 + 14 = 34$

Length = 34cm, Breadth = 6cm

Or

Length = 20+x, Breadth = 20-x **x = 14**

(20+x) (20-x) = 204 Length = 20 + 14 = 34 cm, Breadth = 6 cm

$400 - x^2 = 204$

$x^2 = 400 - 204 = 196$